

6. (25 points)

Blocks a、b、c、d, of mass m, lie on a smooth level table. A spring of natural length l joins a、b; its stiffness is k_1. A spring of natural length l, stiffness k_2, joins c、d. All four centres lie on a line. b、c collisions are perfectly elastic and very brief. Both springs initially have natural length; c、d rest, while a、b move right at the same velocity v_0. Discuss quantitatively:

(1) For $k_1=k_2$, how do the four blocks move relative to the table?

(2) For $k_1=4k_2$, how do the four blocks move relative to the table?